Sai Veeresh Bandi

CFD engineer

About me

I specialize in computational fluid dynamics, with a strong focus on compressible and hypersonic flows (Mach 2–20), where shock waves, expansions, and thermal effects dominate.

Trained through Flow ThermoLabs comprehensive CFD courses (ANSYS Fluent to advanced level and foundational numerics), I build stable, physically accurate simulations using ANSYS Fluent and OpenFOAM.

My emphasis is on solver strategy, high-quality meshing, turbulence modeling (k-ω SST), and rigorous validation to deliver reliable results in high-speed regimes.

Driven by propulsion and ambitious aerospace systems—from deep-space exploration (e.g., JPL) to fast-iterating launch startups—I thrive on engineering at the edge, curiosity in the details, and always validating.

Skills & expertise

CFD & Physics

- Compressible & hypersonic flows (Mach 2–20)
- Shock wave & expansion analysis
- Boundary layer behavior in high-speed flows
- Turbulence modeling (k-ω SST)
- Species transport & reacting flows

Tools

- ANSYS Fluent
- OpenFOAM
- MATLAB
- Python (post-processing)
- SolidWorks, CATIA, HyperMesh

Numerical Methods

- Density-based & pressure-based solvers
- AUSM schemes
- Mesh independence & domain independence
- Convergence & residual analysis

Professional Experience

Design Engineer

HRBL Group | November 2024 – Present (Delhi, India)

- Designed and analyzed electric vehicle (EV) components, including chassis, battery housings, cooling systems, and structural mounts using SolidWorks and CATIA.
- Led structural, modal, thermal, and CFD-driven analyses in ANSYS to optimize durability, thermal management, and cooling efficiency for EV subsystems.
- Conducted CFD thermal simulations for battery and motor cooling, applying reverse engineering to upgrade legacy designs and reduce costs.
- Performed multiphysics validation (structural + thermal + fluid) ensuring compliance with performance and safety standards.

Product Development Intern

Simulation Lab | September 2024 – October 2024

- Conducted aerodynamic CFD analysis on subsonic aircraft wing geometries using ANSYS Fluent.
- Computed lift/drag coefficients, refined meshes, and optimized boundary conditions for accurate force predictions.

Aerospike Nozzle

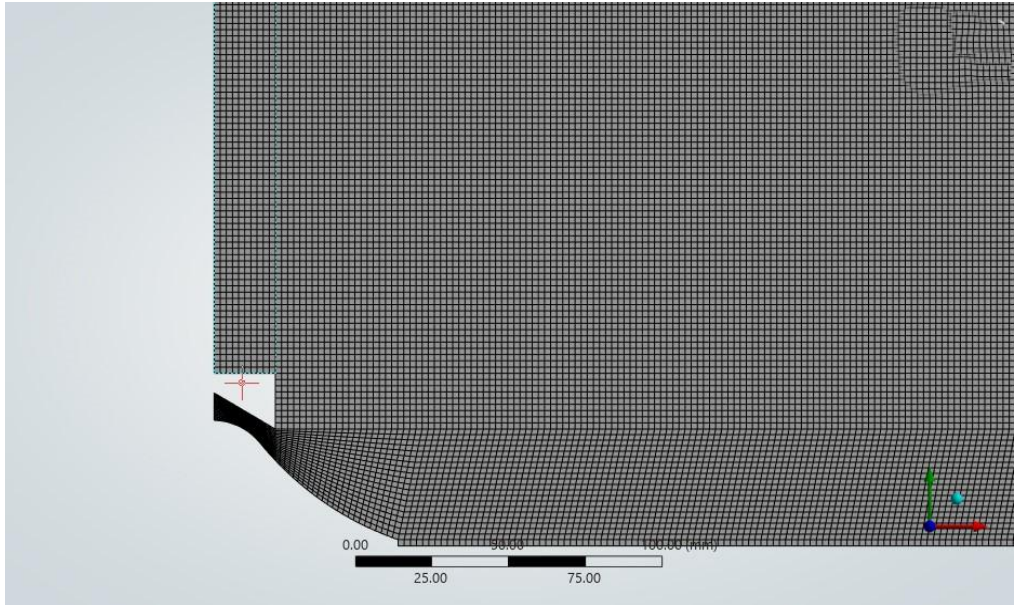
A modified 2d axisymmetric aerospike was analyzed and to study the expansion behaviour using compressible flow .

Basic nozzle relations were used to estimate flow parameters, and the geometry was adjusted before performing CFD simulations.



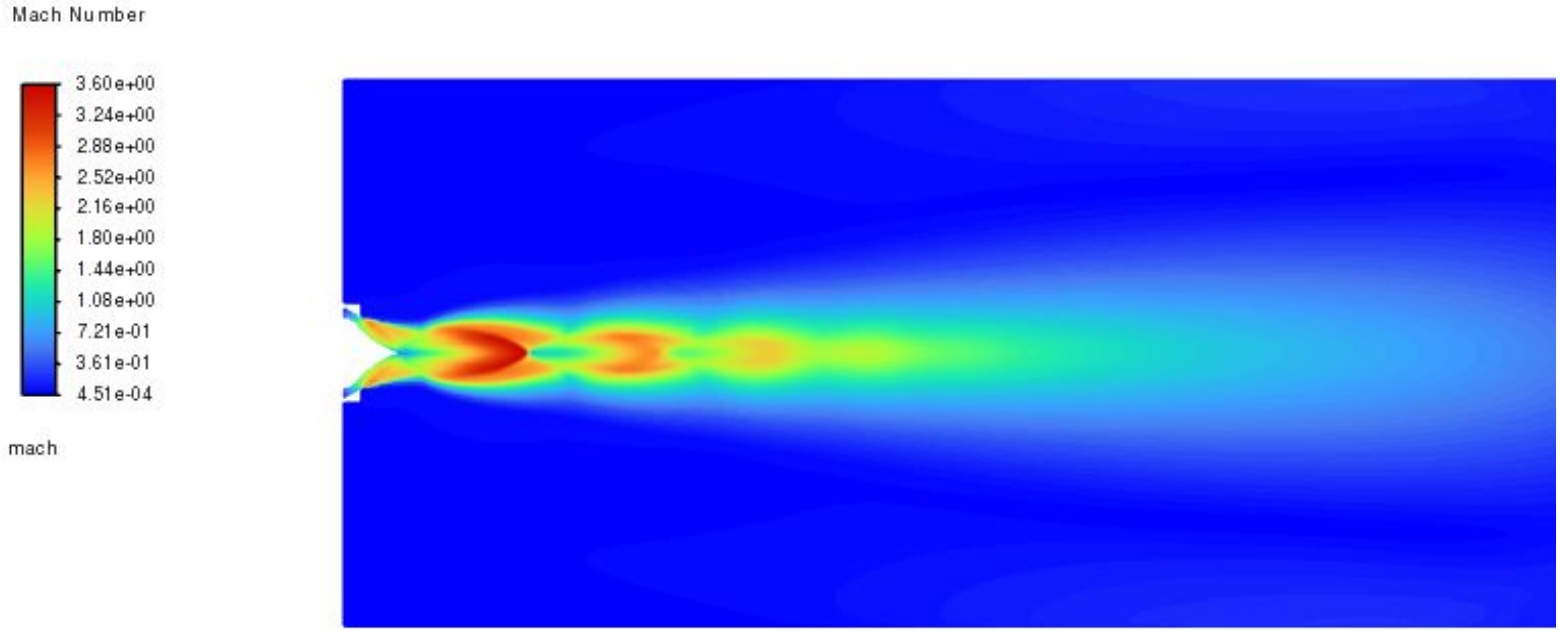
Setup

1. Solver: Density Based
2. Operating Pressure: 0Pa
3. Energy equation: on
4. K-omega SST
5. Fluid: Ideal gas air
6. Viscosity : sutherland
7. Pressure Inlet: 15.2bar
8. Temperature inlet: 1180K
9. Pressure outlet: 101325 Pa

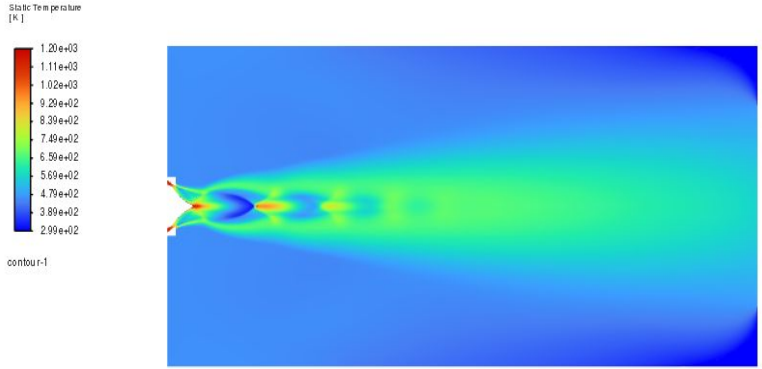


Mesh

1. Element size: 2mm
2. Number of nodes: 108274
3. Number of elements: 107402
4. Face Meshing & quad dominant method

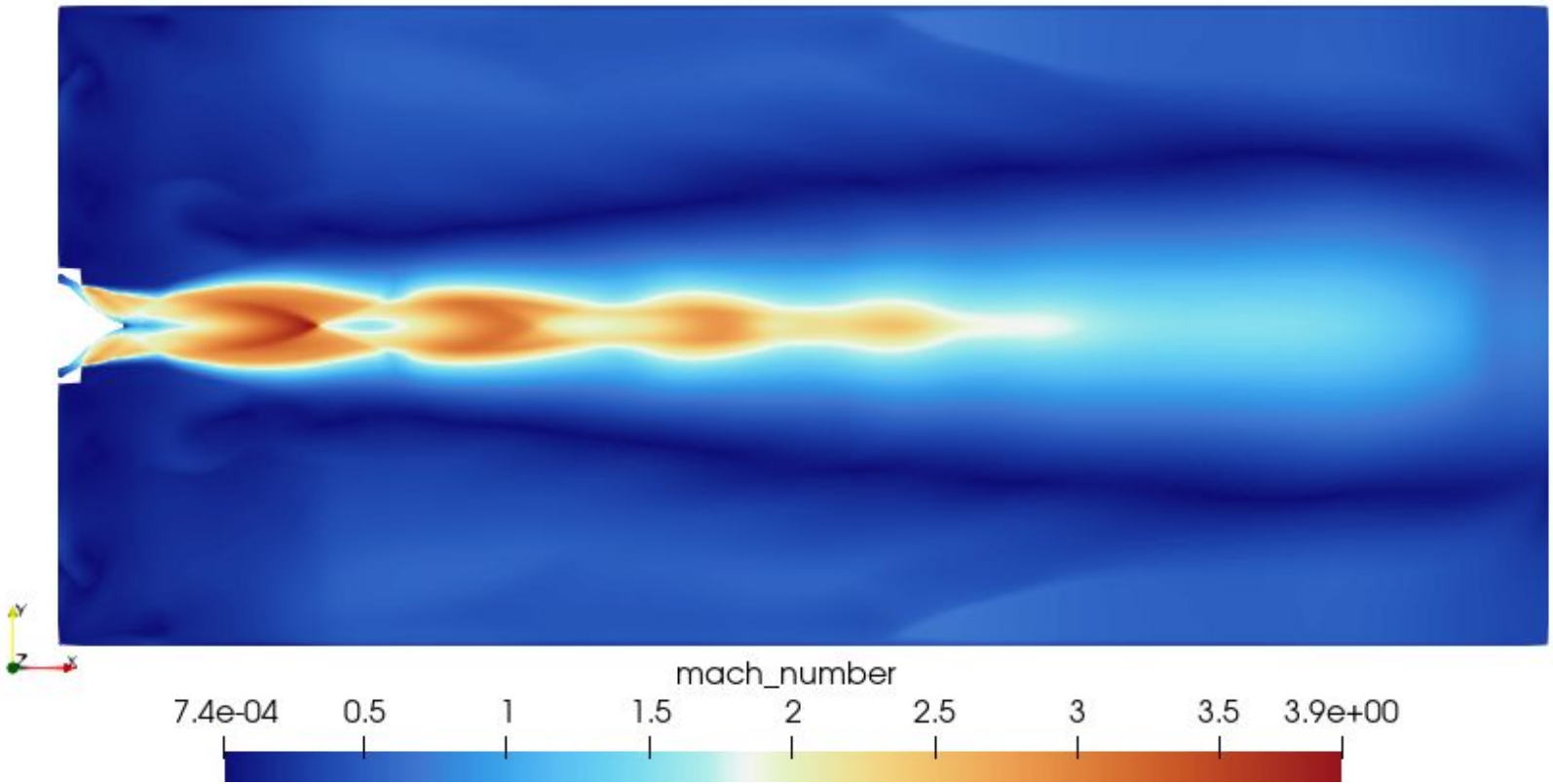


The CFD results demonstrate the formation of shock diamonds caused by pressure imbalance between the nozzle exit and ambient conditions. The Mach contours show periodic expansion and recompression waves, while the temperature field reflects the thermodynamic effects of compressible flow expansion.



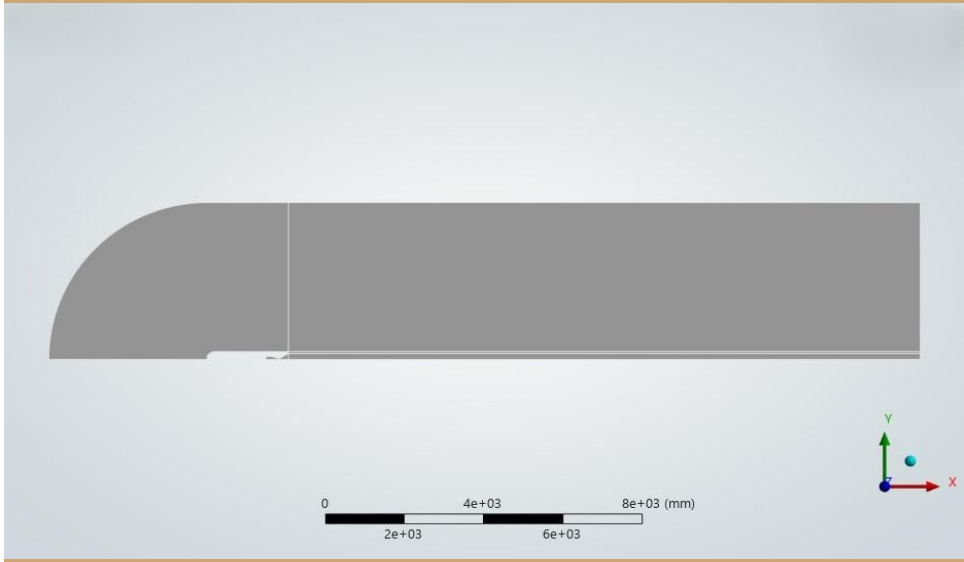
Mesh convergence study

	Element size	nodes	elements	mach
Coarse Mesh	4mm	25299	24870	2.67
Medium Mesh	2mm	108274	107402	3.60
Fine mesh	0.9mm	486475	484580	3.90


Y
Z
X
mach_number
7.4e-04
0.5
1
1.5
2
2.5
3
3.5
3.9e+00

Coupled aerodynamics & Nozzle

Pressure based
vs
density based solver



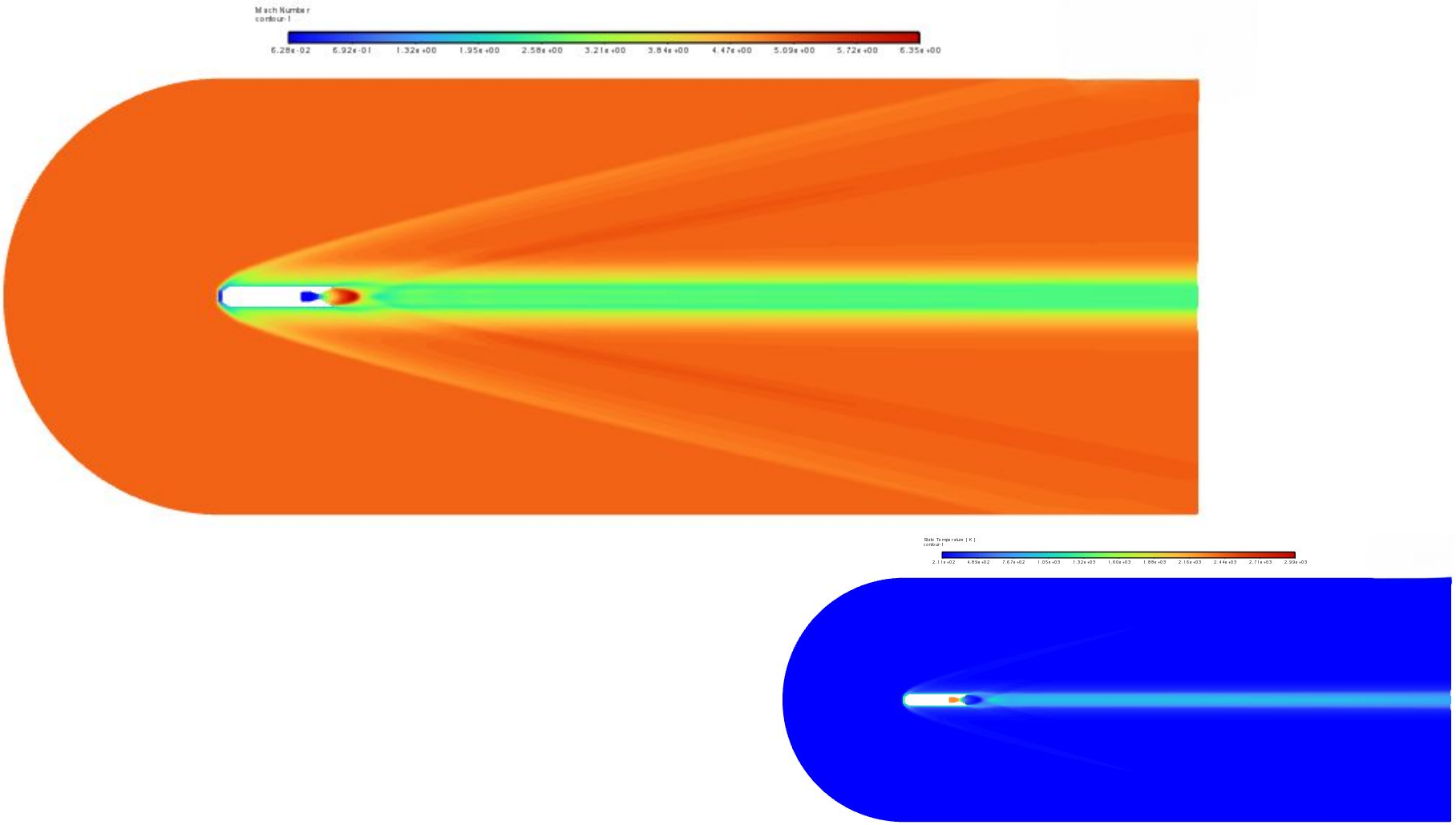
Mesh

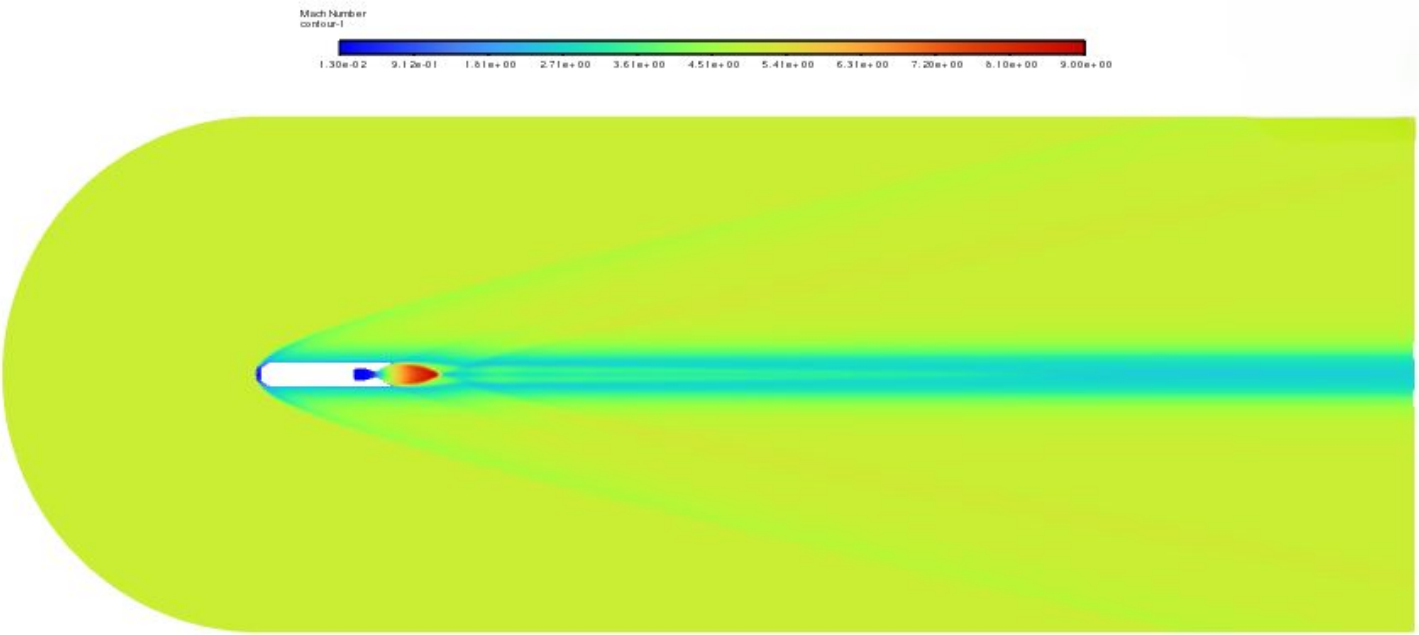
1. Element size : 25mm
2. Number of nodes : 138 946
3. Number of elements: 137926
4. Inflation: first layer thickness: 1mm
5. Number of layers: 15
6. Growth:1.5

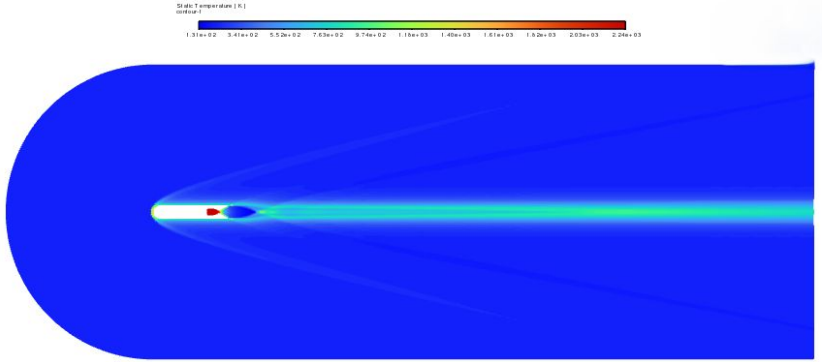
Setup

1. Pressure based & density based
2. Energy equation: on
3. K omega SST
4. Fluid: ideal gas -air
5. Viscosity: sutherland
6. Free stream pressure : 5400Pa
7. Free stream mach : 5
8. Free stream temperature: 220 Kelvin
9. Pressure inlet: 14.7e5 Pa
10. Temperature inlet: 2240 kelvin
11. Pressure outlet same as free stream

This project investigates the jet plume interaction with the aerodynamic freeflow. The objective was to observe how the high-speed exhaust plume interacts with the surrounding flow field and influences the local pressure and velocity distribution. The study was made in the both density and pressure based solvers


Mach Number
contour-1
6.28e-02
6.92e-01
1.32e+00
1.95e+00
2.58e+00
3.21e+00
3.84e+00
4.47e+00
5.09e+00
5.72e+00
6.35e+00
Static Temperature [K]
contour-1
2.11e+02
4.89e+02
7.67e+02
1.05e+03
1.32e+03
1.60e+03
1.88e+03
2.16e+03
2.44e+03
2.71e+03
2.99e+03


Mach Number
contour-1
1.30e-02
9.12e-01
1.81e+00
2.71e+00
3.61e+00
4.51e+00
5.41e+00
6.31e+00
7.20e+00
8.10e+00
9.00e+00

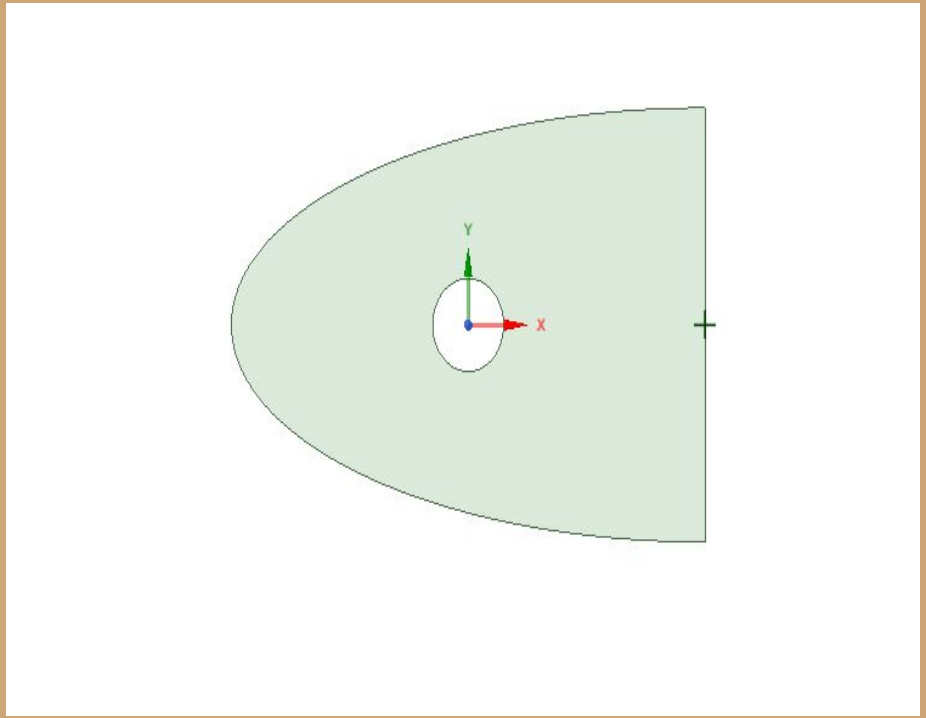

Static Temperature [K]
contour-1
1.31e+02
3.41e+02
5.52e+02
7.63e+02
9.74e+02
1.18e+03
1.40e+03
1.61e+03
1.82e+03
2.03e+03
2.24e+03

A comparative study between pressure-based and density-based solvers was conducted for the compressible flow case.

- The pressure-based coupled scheme handled the pressure–velocity coupling efficiently, leading to faster stabilization of the solution
- The pressure-based solver converged faster, achieving a mass imbalance of approximately **0.001% with fewer iterations**, indicating improved numerical stability and efficient convergence for this configuration
- The shock and expansion structures were captured similarly in both cases, confirming that both numerical approaches are capable of resolving the supersonic flow physics.

Rotating cylinder
In
Mach 6 flow

Transient Hypersonic Flow Over Rotating Cylinder

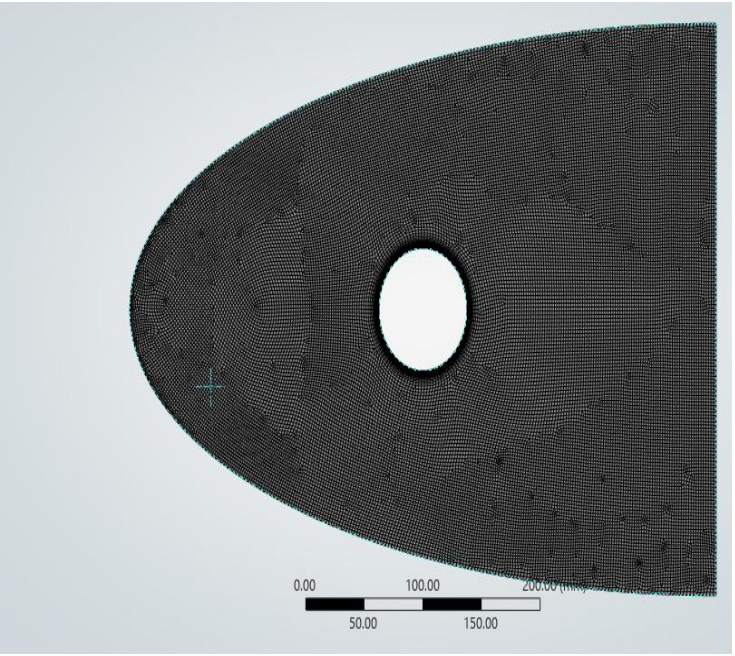


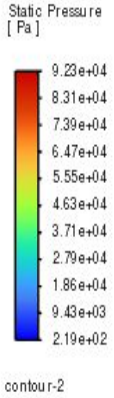
At the center of the domain, a 75 mm diameter cylinder was modeled with a rotational speed of 1074 rad/s. The objective of the study was to **analyze how hypersonic flow interacts with a rotating body and how rotation influences shock structure, wake formation, and local flow acceleration.**

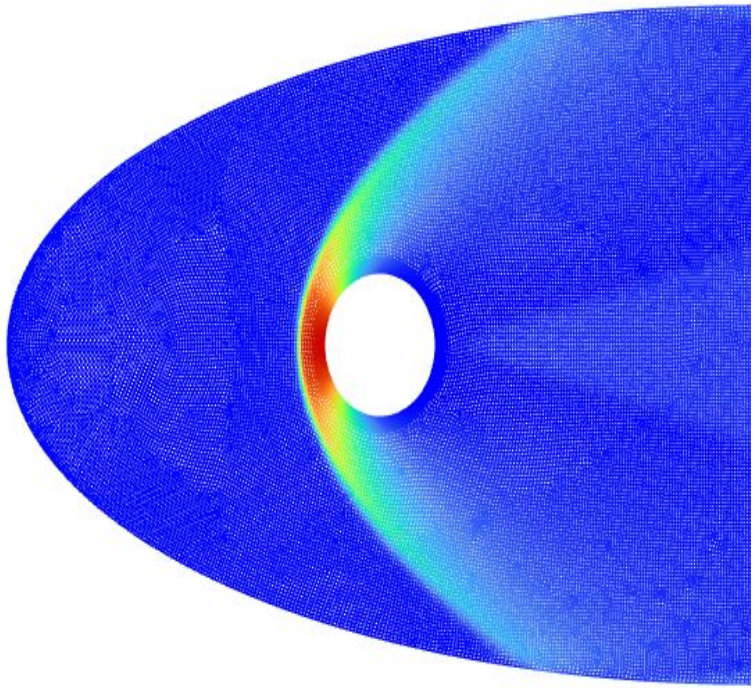
1. Element size: 2mm
2. No of nodes: 34472
3. No. of elements: 34079
4. Quad dominant mesh
5. Inflation with first layer of thickness : 0.05mm
6. No. of layers: 20
7. Growth: 1.2

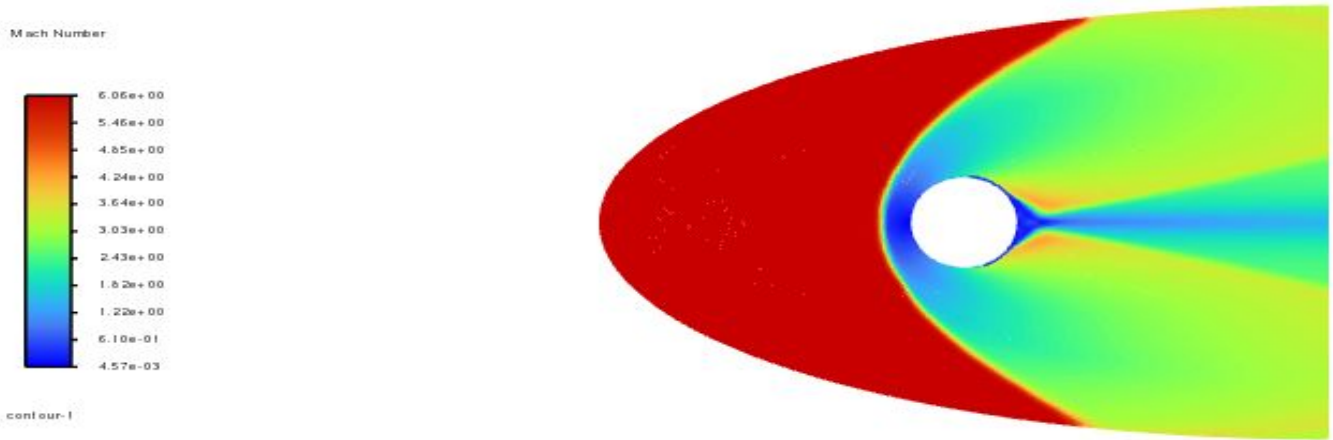
Setup

1. Density based solver
2. Energy equation: on
3. K omega SST
4. FLuid: ideal gas - air
5. Viscosity : sutherland
6. Free stream pressure : 1951 pa
7. Free stream temperature: 58.25 k
8. Free stream mach 6.06
9. Wall: cylinder wall with rotation of 1074 rad/s
10. Operating pressure: 0 pa
11. Explicit with ausm
12. Transient formulation: first order implicit


0.00
100.00
200.00 (mm)
50.00
150.00


Static Pressure
[Pa]
9.23e+04
8.31e+04
7.39e+04
6.47e+04
5.55e+04
4.63e+04
3.71e+04
2.79e+04
1.86e+04
9.43e+03
2.19e+02
contour-2



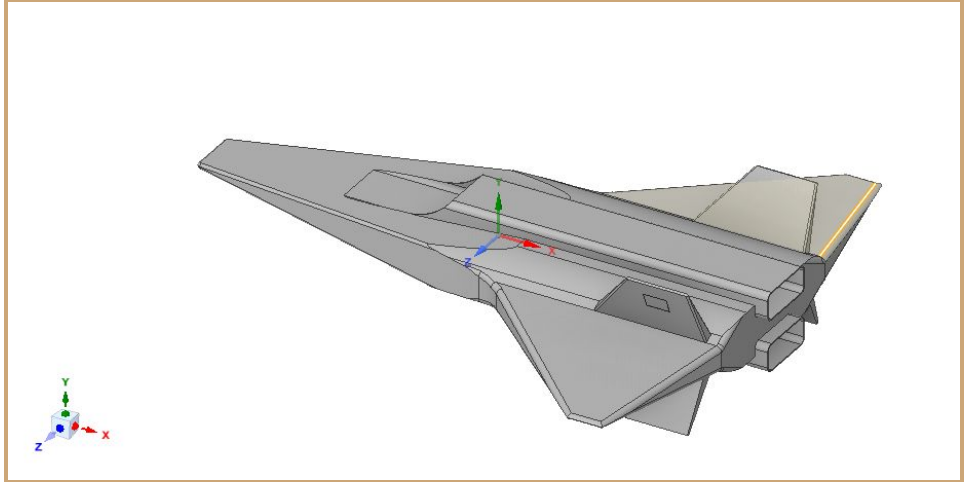


The simulation shows

- the formation of a detached bow shock at a finite distance ahead of the rotating cylinder under Mach 6 freestream conditions.
- Due to the high rotational speed (1074 rad/s), the flow around the cylinder becomes asymmetric, modifying the pressure and velocity distribution along the surface..

NASA X-43 Hypersonic Flow Analysis

Mach-7 flow
Frozen chemistry
Vs
Species



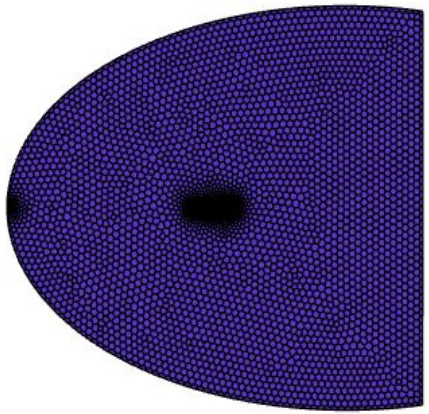
Baseline geometry sourced from GrabCAD and refined for computational analysis, including mesh preparation and domain setup.

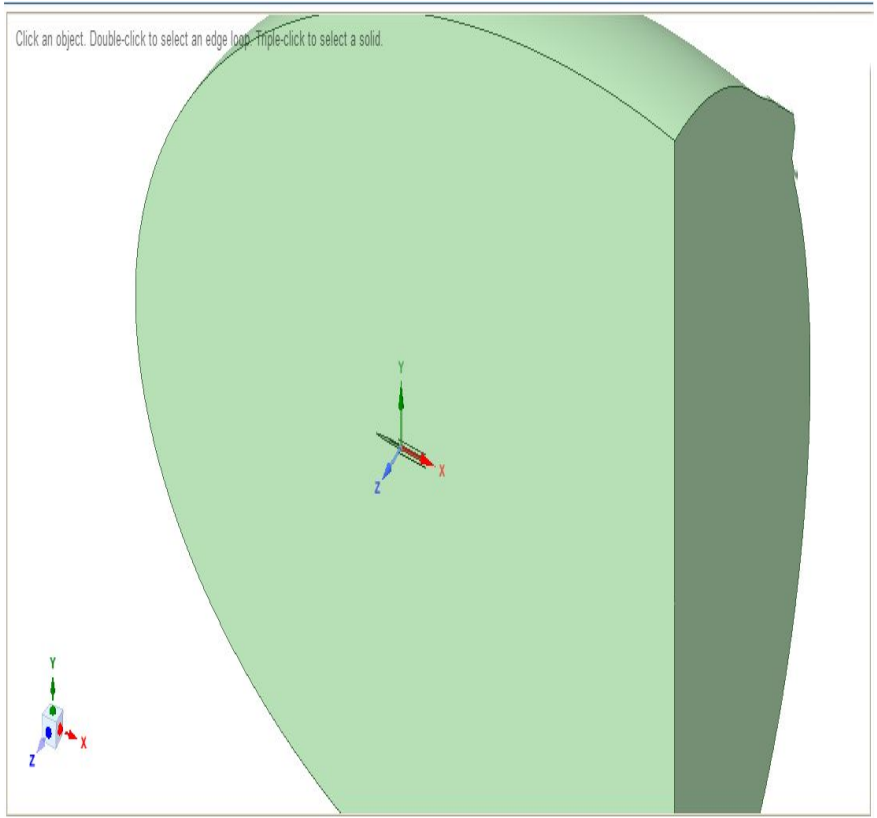
Description

To better capture high-temperature gas effects typical of hypersonic flows, the simulation incorporated:

- **Frozen chemistry modeling**
- **Air species composition**

These models allow the solver to account for the thermodynamic behavior of air at high Mach numbers, where compressibility and temperature effects become significant.



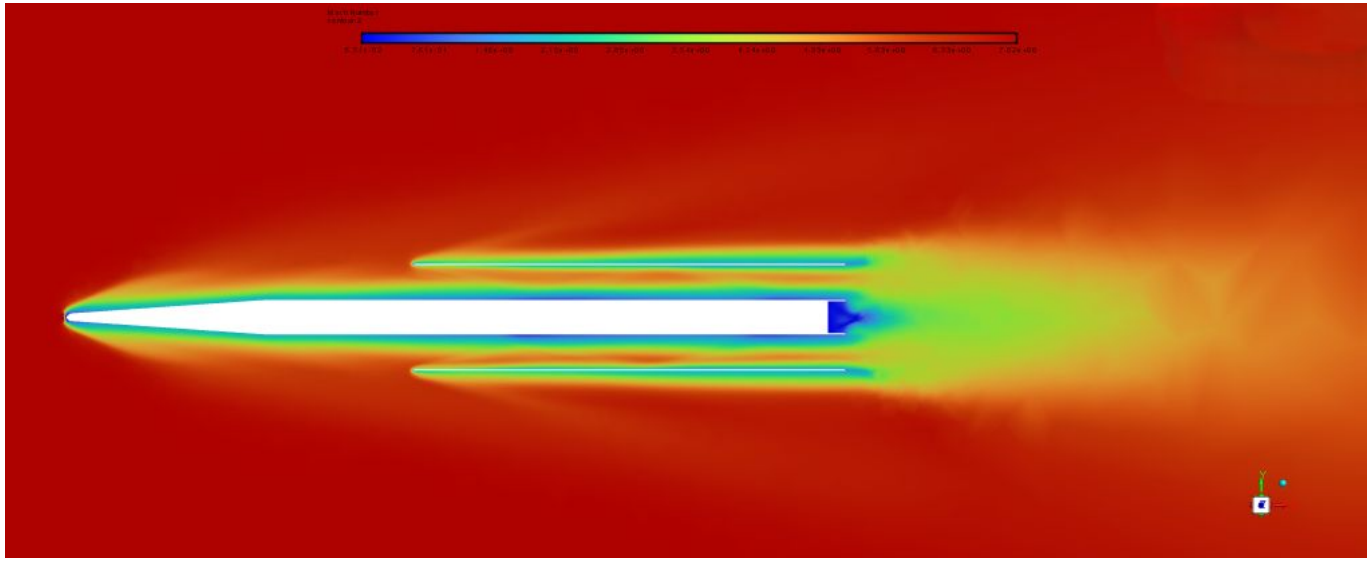

Click an object. Double-click to select an edge loop. Triple-click to select a solid.
Y
X
Z
Y
X
Z

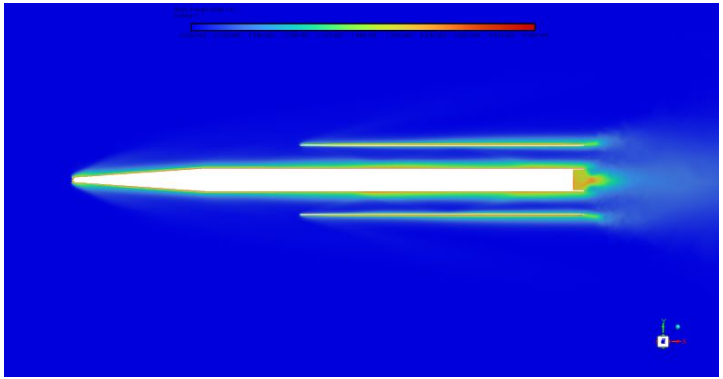
Mesh

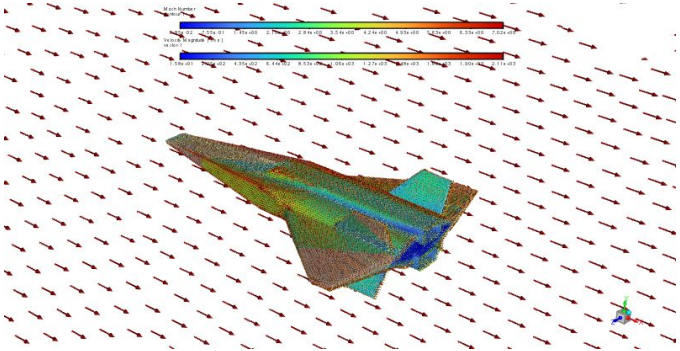
	Boundary	interior
Nodes	101086	795465
faces	32771	1210469
total	243529	
Shape	Polyhedral with a size of 2.5m	
Last ratio	0.05m with 15 layers	
Walls element size	0.09m	

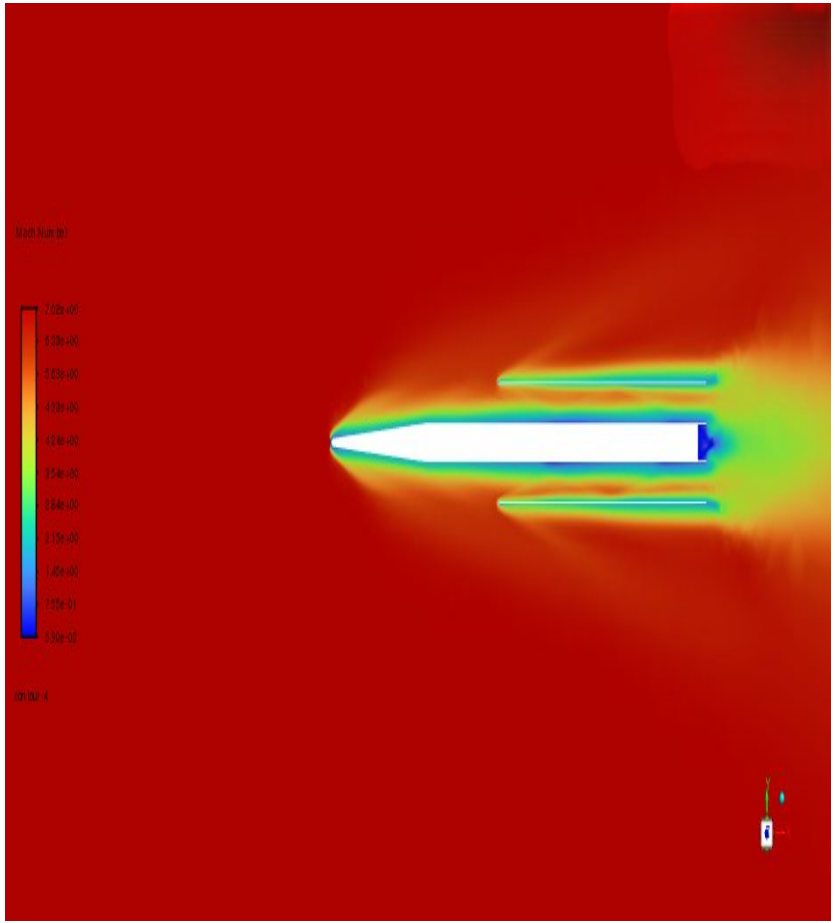
Setup

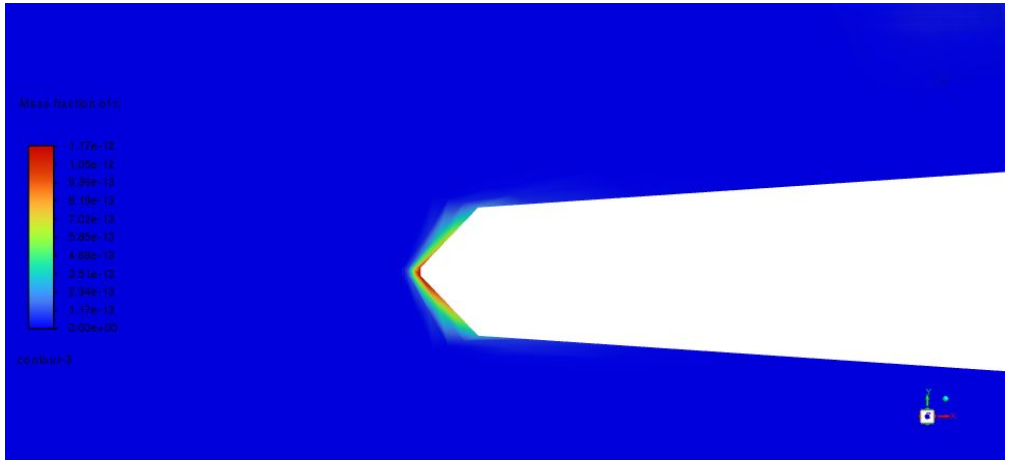
1. Density based solver
2. Energy equation : on
3. Symmetric model
4. Fluid: air -ideal gas
5. Cp : NASA 9-piecewise polynomial
6. Viscosity : sutherland
7. Free stream pressure: 1400 pa
8. Free stream mach: 7
9. Free stream temperature: 226 k
10. Free stream species O2: 0.21%
11. Species taken: nitrogen, oxygen,

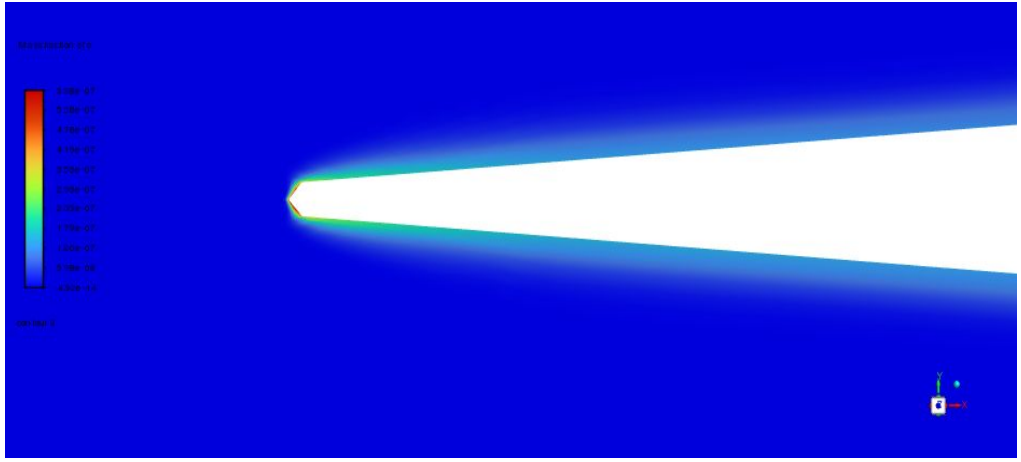







Mach Number
7.02e+00
6.33e+00
5.63e+00
4.93e+00
4.24e+00
3.54e+00
2.84e+00
2.15e+00
1.45e+00
7.57e-01
5.90e-02
contour 4


Mass fraction of n
1.17e-12
1.05e-12
9.36e-13
8.19e-13
7.02e-13
5.85e-13
4.68e-13
3.51e-13
2.34e-13
1.17e-13
0.00e+00
contour 3


contour 2

Parameter	CFD	Test results
Drag Coefficient (Cd)	0.01662601	0.015 -0.018
Pressure Drag %	~93%	85–95%
Viscous Drag %	~7%	5–15%
Forces calculated	66,991.68 N	
Flow Regime	Pressure -dominated	Pressure -dominated

Results are compared with published hypersonic aerodynamic data, showing good agreement in drag coefficient and flow physics.

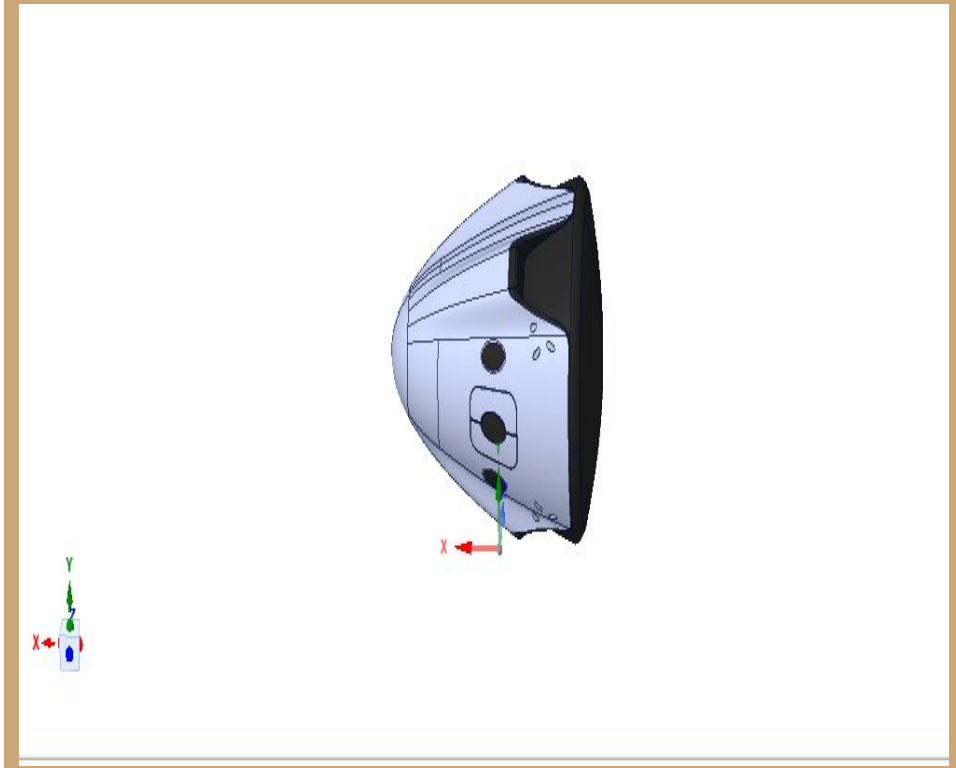
References :

[1] À. Navó and J. M. Bergadà, "Aerodynamic Study of the NASA's X-43A Hypersonic Aircraft," Appl. Sci., vol. 10, no. 22, p. 8211, 2020, doi: 10.3390/app10228211.

[2] M. C. Davis and J. T. White, "X-43A Flight-Test-Determined Aerodynamic Force and Moment Characteristics at Mach 7.0," J. Spacecr. Rockets, vol. 45, no. 3, pp. 472–484, 2008, doi: 10.2514/1.30413.

SpaceX dragon capsule

Hypersonic Flow Over Re-entry Capsule



Baseline geometry sourced from GrabCAD and refined for computational analysis, including mesh preparation and domain setup.

The objective of the simulation is to observe:

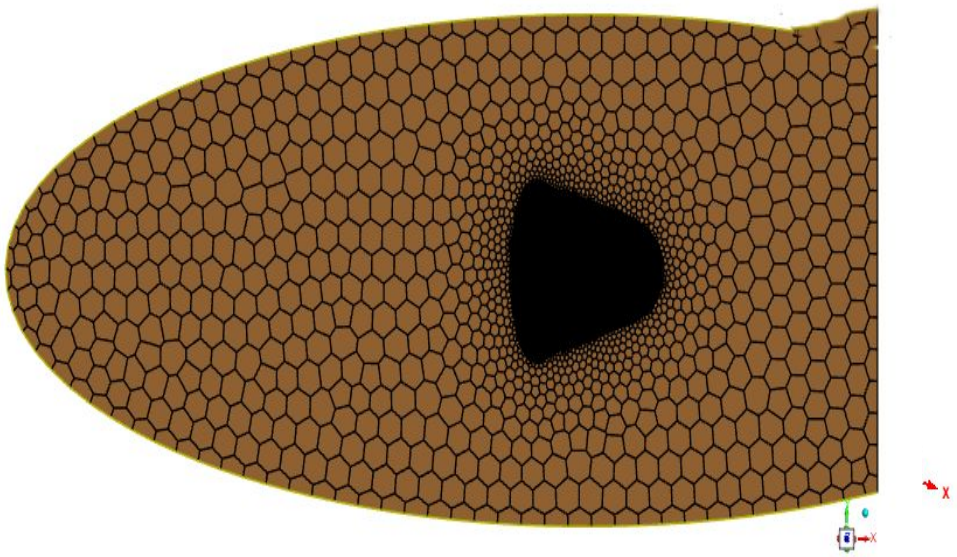
- **Bow shock formation ahead of the blunt body**
- **Flow deceleration and compression in the shock layer**
- **Pressure and temperature distribution over the capsule surface**
- **Wake structure behind the capsule**

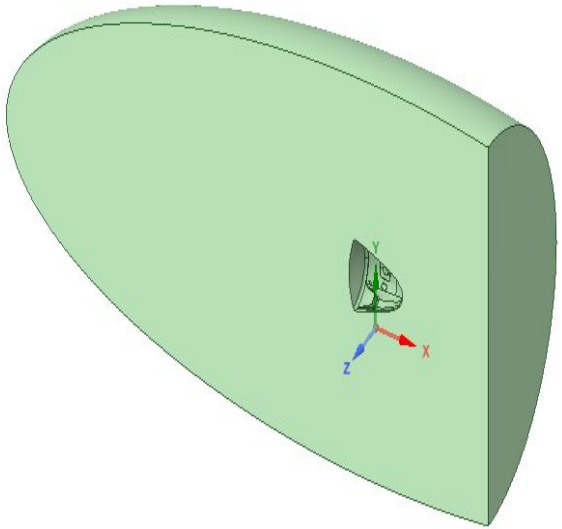
Setup

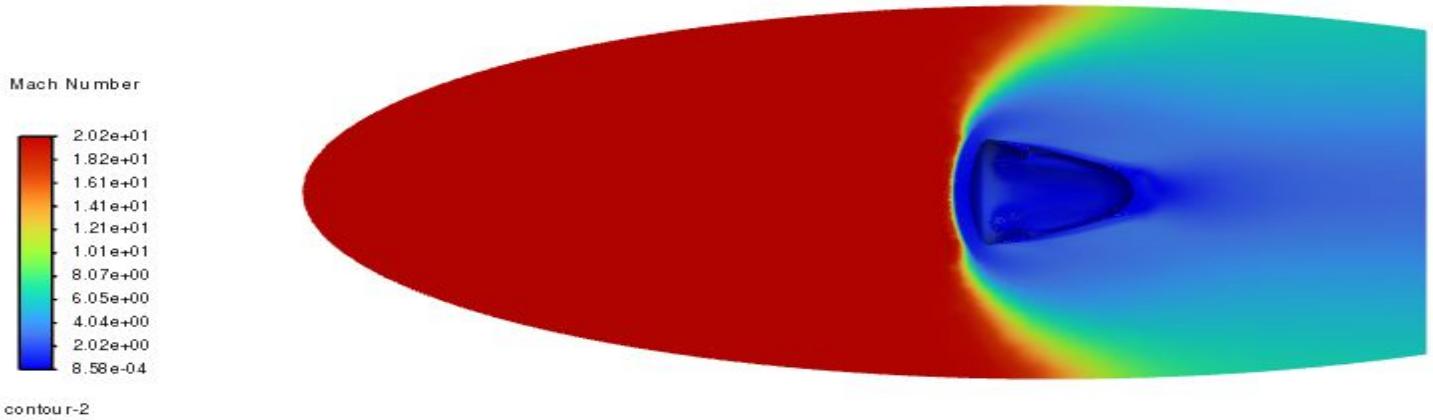
1. Density based
2. Energy equation : on
3. Laminar
4. Fluid : ideal gas air
5. Cp - nasa 9 piecewise polynomial
6. Viscosity : sutherland
7. Freestream pressure: 50Pa
8. free stream temperature : 216 K
9. Freestream mach : 20
10. Wall temperature: 1000k

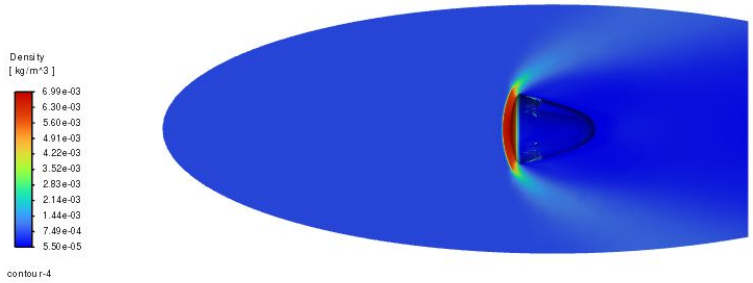
Mesh

	Boundary	interior
nodes	112498	1134777
faces	32917	1920491
	426035	
Face size on walls	0.035 m	
Boundary layers	First thickness 0.03m with 15 layers	


X
X


Y
X
Z


Mach Number
2.02e+01
1.82e+01
1.61e+01
1.41e+01
1.21e+01
1.01e+01
8.07e+00
6.05e+00
4.04e+00
2.02e+00
8.58e-04
contour-2


Density
[kg/m^3]
6.99e-03
6.30e-03
5.60e-03
4.91e-03
4.22e-03
3.52e-03
2.83e-03
2.14e-03
1.44e-03
7.49e-04
5.50e-05
contour-4

The simulation results demonstrate the formation of a

- **strong detached bow shock ahead of the Dragon capsule**, a typical characteristic of blunt-body re-entry vehicles operating in hypersonic regimes.
- At **Mach 20**, the incoming flow undergoes rapid compression across the shock wave, leading to a substantial rise in temperature and pressure near the stagnation point.
- The blunt geometry effectively decelerates the hypersonic flow, creating a thick shock layer between the capsule surface and the bow shock, which helps reduce the heat transfer to the vehicle structure.

Parameter	Your CFD	Published Data
Drag Coefficient	**1.26**	1.23
Drag Type	Pressure-dominated	Pressure-dominated
Forces CFD	104,213.72 N	
Flow Physics	Shock-driven	Shock-driven

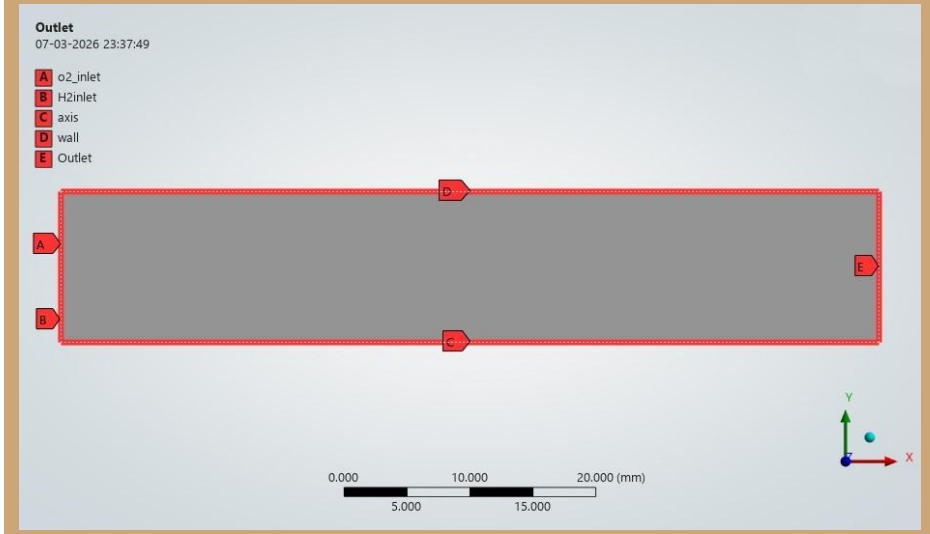
The computed Cd = 1.26 shows excellent agreement with published values of Cd = 1.23 derived from NASA Ames CFD and ballistic coefficient analysis for Crew Dragon re-entry configuration. The strong detached bow shock, stagnation-point heating, and pressure-dominated drag are physically consistent with blunt-body hypersonic aerodynamics."

[1] A. A. Gonzales et al., "Mars Sample Return using commercial capabilities: Mission architecture overview," 2014 IEEE Aerospace Conference, pp. 1-15, 2014,

doi: 10.1109/AERO.2014.6836421.

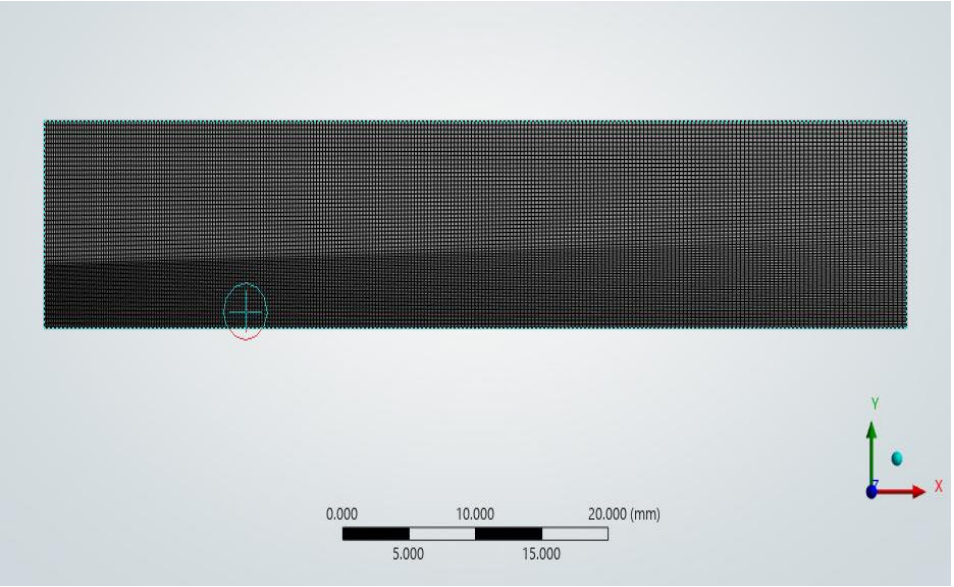
Combustion

Oxygen+ hydrogen



This project investigates the combustion behavior of liquid oxygen (LOX) and liquid hydrogen (LH_2) in a rocket propulsion environment. The objective is to study the mixing, chemical reaction, and temperature distribution resulting from the combustion of these cryogenic propellants.

The propellant combination of Liquid Hydrogen and Liquid Oxygen is widely used in high-performance rocket engines due to its high specific impulse and clean combustion products. This propellant pair is used in several launch vehicles and space propulsion systems, including engines developed by NASA

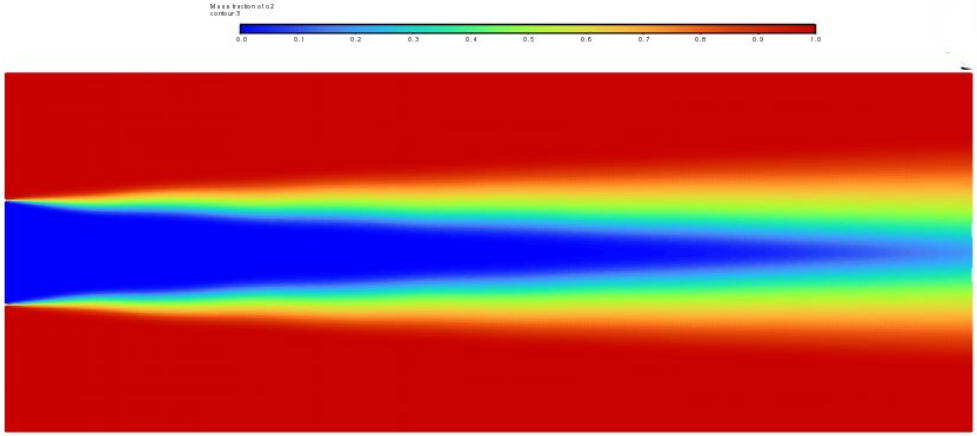


Mesh

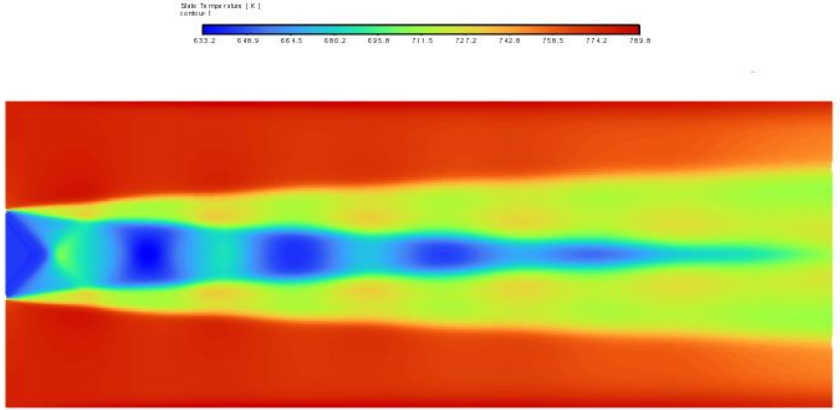
1. No of nodes : 17487
2. No of elements: 17160
3. Quad mesh
4. Element size : 3mm

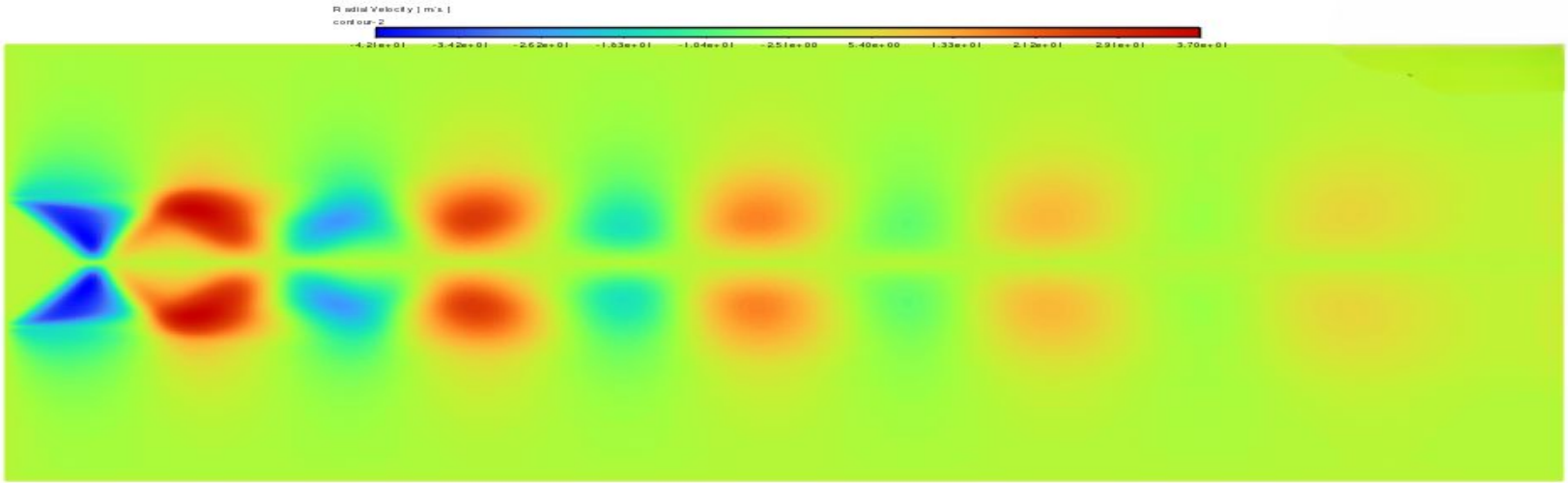
Setup

1. Pressure based
2. Energy equation : on
3. Species: H2- O2
4. Hydrogen inlet: 0.0126 kg/s
5. Hydrogen temperature: 858 K
6. Oxygen inlet: 0.1 kg/s
7. Oxygen pressure: 1e6 pa
8. Oxygen temperature : 858K
9. Operating pressure: 101325 pa



Overall, the results demonstrate that efficient mixing between hydrogen and oxygen is the key factor governing combustion intensity and temperature rise in cryogenic rocket propellants, highlighting the importance of injector design and flow dynamics in high-performance rocket engines

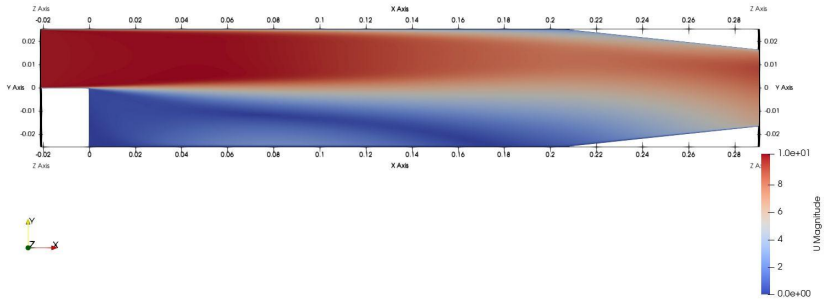


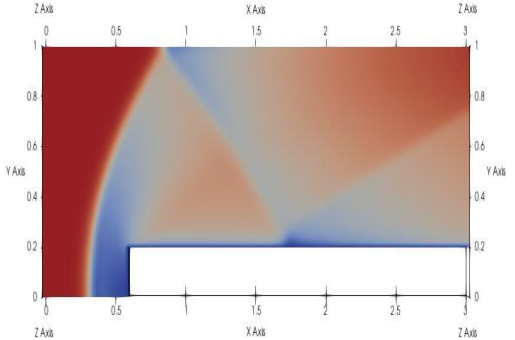


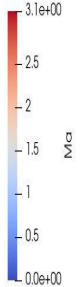
The simulation illustrates the mixing and combustion characteristics of liquid oxygen and liquid hydrogen in a reacting flow environment. The species fraction contours show the progressive mixing of oxidizer and fuel streams, forming a reaction zone downstream of the injector region.

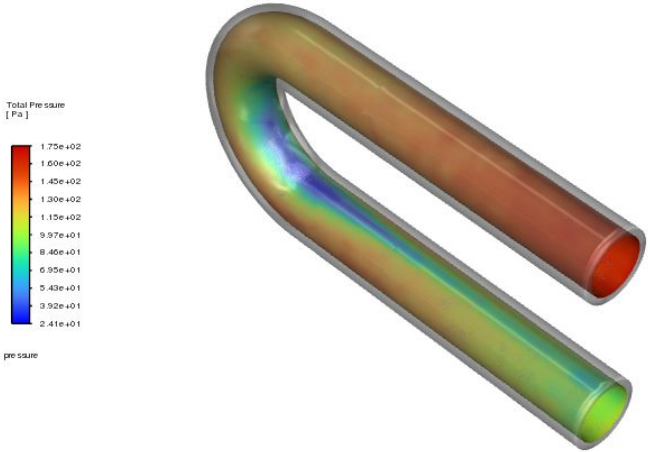
The radial velocity contours reveal the presence of alternating vortical structures and flow oscillations, which enhance turbulent mixing between hydrogen and oxygen. These vortical motions improve reactant interaction and promote efficient combustion along the flow direction.

Other works

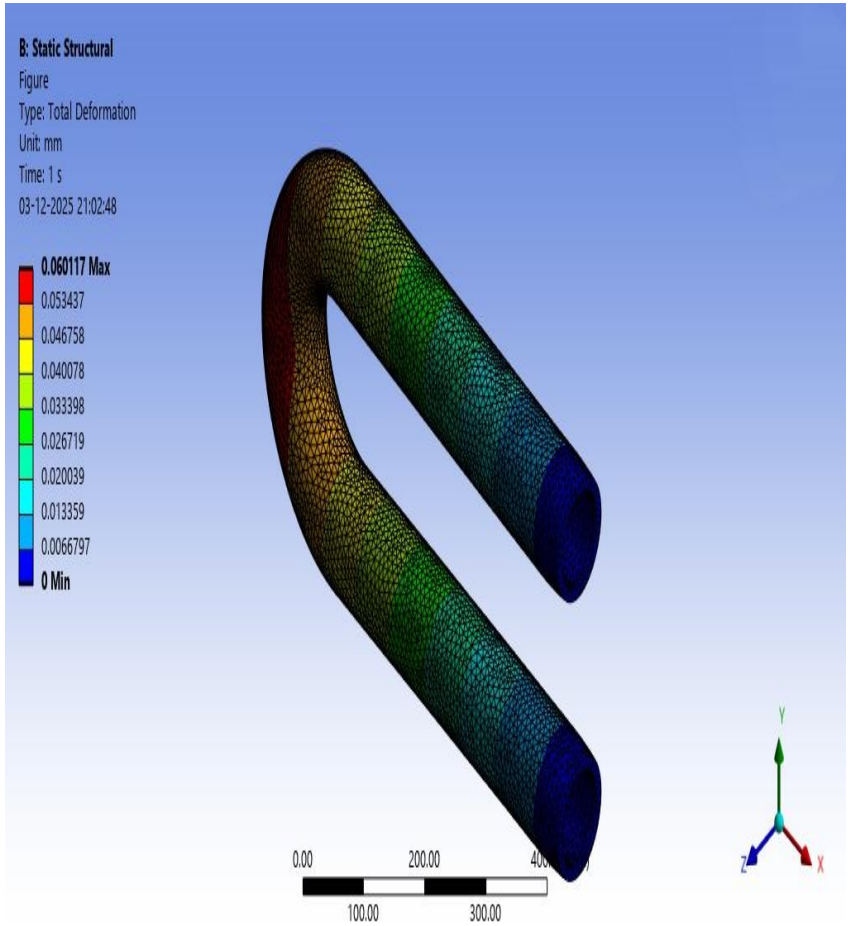

X Axis
Y Axis
Z Axis
U Magnitude
1.0e+01
8
6
4
2
0.0e+00

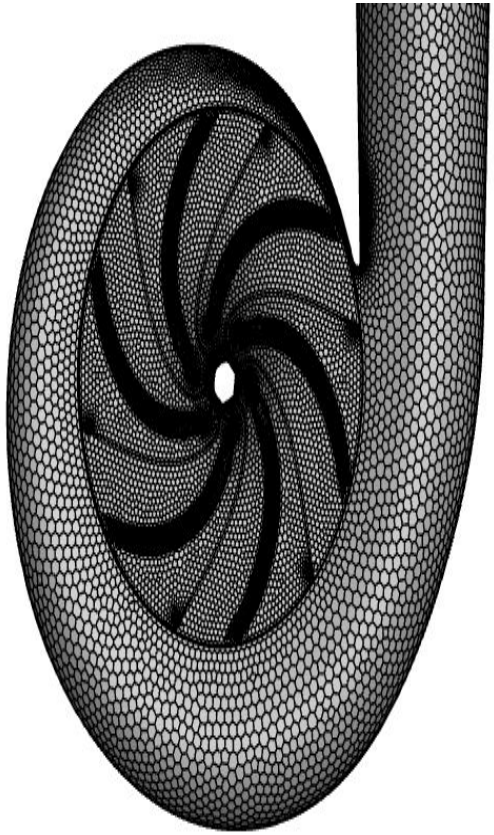

X Axis
Y Axis
Z Axis

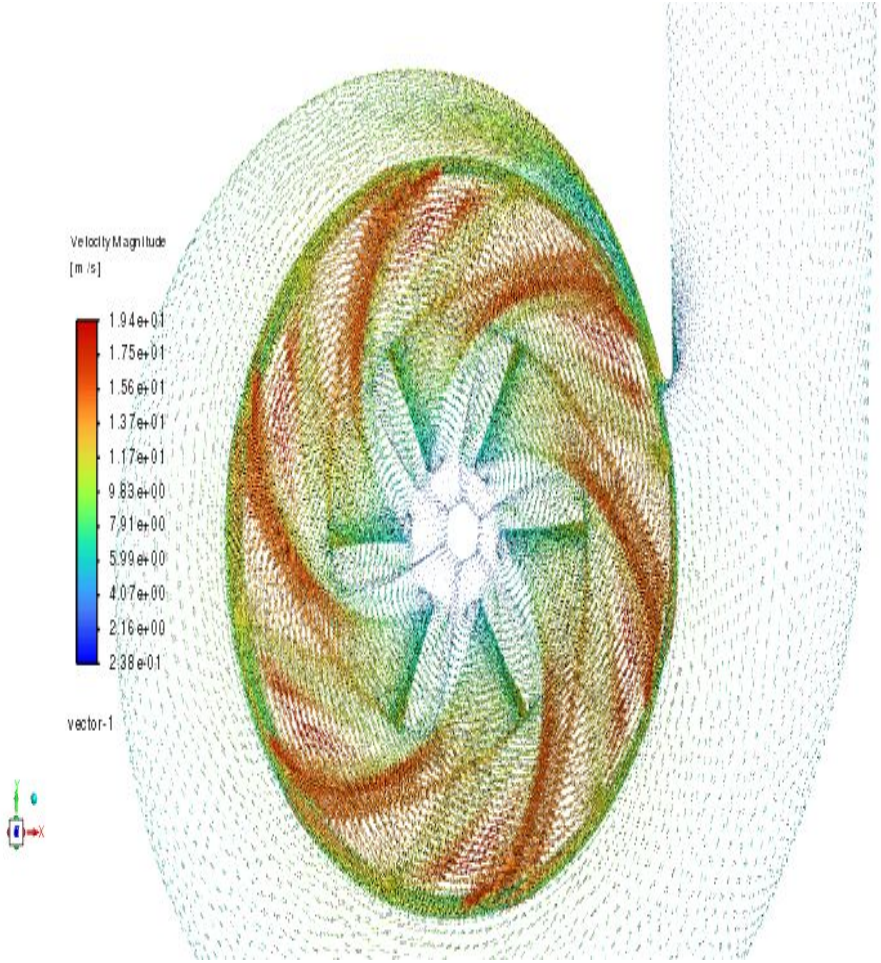

3.1e+00
2.5
2
1.5
1
0.5
0.0e+00
Ma


Total Pressure
[Pa]
1.75e+02
1.60e+02
1.45e+02
1.30e+02
1.15e+02
9.97e+01
8.46e+01
6.95e+01
5.43e+01
3.92e+01
2.41e+01
pressure


Velocity Magnitude
[m/s]
2.63e+01
2.45e+01
2.26e+01
2.08e+01
1.90e+01
1.71e+01
1.53e+01
1.34e+01
1.16e+01
9.76e+00
7.91e+00
vector 1


B: Static Structural
Figure
Type: Total Deformation
Unit: mm
Time: 1 s
03-12-2025 21:02:48
0.060117 Max
0.053437
0.046758
0.040078
0.033398
0.026719
0.020039
0.013359
0.0066797
0 Min
0.00
200.00
100.00
300.00
Y
Z
X




Velocity Magnitude
[m/s]
1.94e+01
1.75e+01
1.56e+01
1.37e+01
1.17e+01
9.83e+00
7.91e+00
5.99e+00
4.07e+00
2.16e+00
2.38e-01
vector-1

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